

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	ITA0517	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)		
Student Name:	Yoga Madha A	Reg. No.:	192425058
Section / Batch:	B	Date:	19-08-2026

Code of Conduct

/ Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student



Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

21. Noise Impact on Detection

Results:

Noise Level	Accuracy
Low	92%
Medium	80%
High	60%

Compare performance and evaluate system robustness.

Answer:

The performance of the detection system is strongly affected by the level of noise present in the input data. The results show that accuracy decreases as the noise level increases.

Noise Level	Accuracy	Performance Evaluation
Low	92%	Excellent detection performance
Medium	80%	Good but noticeable performance degradation
High	60%	Poor detection performance

Comparison and Analysis

- Low Noise – 92% Accuracy:**
Under low-noise conditions, the system achieves **92% accuracy**, which indicates that the detection model can identify objects or features reliably when the input data is relatively clean. Important visual features are preserved, allowing the model to make accurate predictions.
- Medium Noise – 80% Accuracy:**
When the noise level increases to medium, accuracy decreases to **80%**. This represents a **12 percentage-point reduction** compared with the low-noise condition. Noise begins to distort important features, making detection more difficult.
- High Noise – 60% Accuracy:**
Under high-noise conditions, accuracy drops significantly to **60%**. Compared with low noise, this is a **32 percentage-point decrease**. This indicates that the model is sensitive to severe disturbances and may produce incorrect detections or miss important objects.
- Overall Trend:**
The results demonstrate a clear **inverse relationship between noise and detection accuracy**. As noise increases, the quality of the input decreases, resulting in poorer detection performance.

Robustness Evaluation

The system can be considered **reasonably robust under low and medium noise**, because it maintains accuracy above 80%. However, its robustness under high-noise conditions is weak because the accuracy falls to only 60%.

To improve robustness, techniques such as **image denoising, data augmentation, noise-aware training, image preprocessing, and robust feature extraction** can be applied.

Conclusion

The system performs best under low-noise conditions with **92% accuracy**. Its performance gradually deteriorates at medium noise and decreases substantially at high noise. Therefore, the system is **effective in relatively clean environments but requires additional noise-handling techniques for reliable operation in highly noisy real-world environments**.

22. Depth Estimation Error Analysis

Results:

Method	Error
A	0.5 m
B	1.2 m
C	0.8 m

Compare and evaluate accuracy for real-world applications.

Answer:

Depth estimation accuracy is evaluated by comparing the estimation error of the three methods. A **lower error indicates better depth estimation accuracy**, because the estimated distance is closer to the actual distance.

Method	Error	Performance Evaluation
A	0.5 m	Best accuracy
B	1.2 m	Lowest accuracy
C	0.8 m	Moderate accuracy

Comparison and Analysis

1. **Method A – 0.5 m Error:**

Method A produces the lowest error of **0.5 m**. This means that its estimated depth is, on average, relatively close to the actual depth. Therefore, Method A provides the **highest accuracy** among the three methods.

2. **Method B – 1.2 m Error:**

Method B has an error of **1.2 m**, which is the highest error. This indicates that its depth predictions deviate considerably from the actual distance. Therefore, Method B has the **weakest accuracy**.

3. **Method C – 0.8 m Error:**

Method C has an error of **0.8 m**. It performs better than Method B but worse than Method A. Hence, it provides **moderate depth estimation accuracy**.

Ranking

The methods can be ranked as:

Method A (0.5 m) > Method C (0.8 m) > Method B (1.2 m)

Here, the smaller error represents better performance.

Real-World Applications

Depth estimation is important in applications such as:

- **Autonomous vehicles:** estimating the distance between vehicles, pedestrians, and obstacles.
- **Robotics:** determining the position of objects for navigation and manipulation.
- **Augmented reality:** placing virtual objects at the correct depth.
- **Surveillance:** estimating the distance of moving objects.
- **3D reconstruction:** generating accurate spatial information from images.

For safety-critical applications such as autonomous driving, even relatively small depth errors can affect decision-making. Therefore, **Method A is the most suitable choice** among the three because it has the lowest error.

Conclusion

Method A is the **most accurate depth estimation method**, with an error of only **0.5 m**. Method C provides acceptable intermediate performance, while Method B has the highest error and is less suitable for applications requiring precise distance estimation. Therefore, **Method A is recommended for real-world applications where depth accuracy is important**.

23. Multi-Model Comparison

Results:

Model	Accuracy	Speed	Robustness
A	88%	Fast	Low
B	92%	Moderate	High
C	95%	Slow	High

Compare and evaluate the best overall model.

Answer:

The three models differ in terms of accuracy, processing speed, and robustness. Therefore, the best overall model should be selected by considering the requirements of the application rather than accuracy alone.

Model	Accuracy	Speed	Robustness	Overall Evaluation
A	88%	Fast	Low	Suitable when speed is the priority
B	92%	Moderate	High	Good balance
C	95%	Slow	High	Highest accuracy but slower

Comparison and Analysis

1. Model A:

Model A achieves **88% accuracy** and has the fastest processing speed. However, its low robustness means that performance may decrease when environmental conditions change. It is suitable for applications where **real-time processing is more important than maximum accuracy**.

2. Model B:

Model B achieves **92% accuracy**, moderate speed, and high robustness. It provides a good compromise between accuracy and computational performance. It is likely to perform consistently under different operating conditions.

3. Model C:

Model C provides the highest accuracy of **95%** and high robustness. However, it is slow compared with Models A and B. This makes it more appropriate for applications where **accuracy is more important than processing speed**.

Best Overall Model

Model B is the best overall model when considering a balance between accuracy, speed, and robustness. Although Model C has higher accuracy, its slow processing can be a disadvantage for real-time applications.

Model B provides:

- High accuracy – **92%**
- Moderate processing speed
- High robustness
- Better balance between performance and computational requirements

Conclusion

Model C is the best choice when **maximum accuracy** is required, while Model A is appropriate when **high speed** is the main requirement. However, **Model B is the best overall model because it provides a balanced combination of accuracy, speed, and robustness**, making it suitable for a wide range of real-world applications.

24. Motion Detection under Crowd Density

Results:

Method	Sparse Crowd	Dense Crowd
Optical Flow	High	Low
Motion Models	Moderate	High

Compare and evaluate the best approach.

Answer:

Motion detection performance changes significantly depending on crowd density. The results show that **Optical Flow performs better in sparse crowds**, whereas **Motion Models perform better in dense crowds**.

Method	Sparse Crowd	Dense Crowd	Evaluation
Optical Flow	High	Low	Good for sparse environments
Motion Models	Moderate	High	Better for dense environments

Comparison and Analysis

1. Optical Flow:

Optical Flow estimates the movement of pixels between consecutive frames. It performs **highly in sparse crowds**, where individual objects have sufficient space and their motion can be clearly identified.

However, in dense crowds, people frequently overlap and move close to one another. This causes difficulties in distinguishing individual motion patterns. Therefore, Optical Flow performance becomes **low in dense crowds**.

2. Motion Models:

Motion Models use learned or predefined patterns of movement to identify motion. They provide **moderate performance in sparse crowds** but achieve **high performance in dense crowds**.

In dense environments, individual movement can be difficult to identify using only pixel-level motion. Motion models can use broader movement patterns and learned representations to handle these complex situations more effectively.

Best Approach

For **sparse crowds**, Optical Flow is a suitable approach because it provides high motion detection performance with relatively direct motion estimation.

For **dense crowds**, Motion Models are the better choice because they maintain **high performance despite occlusion and complex movement patterns**.

Real-World Applications

This analysis is important for:

- Crowd monitoring
- Public transportation stations
- Stadium surveillance
- Shopping malls
- Traffic monitoring
- Public safety systems

Conclusion

There is no single approach that is optimal for every crowd condition. **Optical Flow is best for sparse crowds**, while **Motion Models are best for dense crowds**. For a general-purpose crowd monitoring system, a **hybrid approach combining Optical Flow and Motion Models** could provide better performance across different crowd densities.

25. Model Efficiency vs Accuracy

Results:

Model	Accuracy	Computation
A	90%	High
B	85%	Low
C	88%	Moderate

Compare and evaluate for edge devices.

Answer:

For edge devices, both **accuracy and computational efficiency** are important. Edge devices generally have limited processing power, memory, and energy compared with high-performance servers. Therefore, the model should provide a good balance between accuracy and computation.

Model	Accuracy	Computation	Evaluation
A	90%	High	Highly accurate but computationally expensive
B	85%	Low	Efficient but lower accuracy
C	88%	Moderate	Good balance

Comparison and Analysis

1. Model A – 90% Accuracy, High Computation:

Model A achieves the highest accuracy of **90%**. However, it requires high computational resources. On an edge device, this could result in slower inference, higher energy consumption, and increased memory usage.

2. Model B – 85% Accuracy, Low Computation:

Model B has the lowest accuracy at **85%**, but it requires low computation. This makes it highly efficient for resource-constrained edge devices. It can provide faster inference and lower power consumption.

3. Model C – 88% Accuracy, Moderate Computation:

Model C achieves **88% accuracy** while requiring only moderate computational resources. It offers a compromise between Model A's accuracy and Model B's efficiency.

Suitability for Edge Devices

For edge deployment, the ideal model should not simply have the highest accuracy. It should also have:

- Low computational requirements
- Fast inference
- Low memory consumption
- Low power consumption
- Sufficient prediction accuracy
- Ability to operate without continuous cloud processing

Best Choice

Model C is the best overall choice for edge devices because it provides 88% accuracy with moderate computation.

Conclusion

For edge devices, **Model C is recommended** because it balances accuracy and computational cost. Model A should be selected when maximum accuracy is more important and sufficient hardware resources are available. Model B is suitable when very low computation and power consumption are the primary requirements. Overall, **Model C provides the most practical solution for real-time edge applications.**